

FORMATTING INSTRUCTIONS FOR ICLR 2027

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ABSTRACT

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed as interacting agents that exchange information, negotiate decisions, and coordinate their actions over multiple turns. Such multi-agent systems (MAS) promise to combine information and capabilities distributed across multiple agents, yet recent work has exposed substantial failures that arise specifically through interaction. Agents fail to proactively acquire information that is available only to others (Li et al., 2026a), communicate actively without converting those exchanges into effective distributed reasoning (Zhang et al., 2026b), and remain unable to coordinate effectively even when relevant information is made transparent (Yao et al., 2026). Broader analyses similarly identify inter-agent failures as a distinct source of error beyond mistakes made by individual agents in isolation (Cemri et al., 2025; Anthropic, 2026). These findings raise a basic question: what capabilities are missing when individually capable agents fail to interact effectively?

Classical models of interactive decision making provide a principled way to make this question more precise. In an interactive POMDP, an agent maintains uncertainty not only over the physical environment, but also over models of other agents, and plans its own actions with respect to this interactive state (Gmytrasiewicz & Doshi, 2005). Related work on opponent modeling similarly couples inference about another agent with planning a response to the inferred model (Huang et al., 2024). Viewed through this lens, effective interaction requires at least two computations: **interactive-state inference**, which construct belief of other agents relevant information, and **interaction planning**, which uses that belief to evaluate what to do next. Importantly, these computations are coupled over time. An inferred state changes the value of a interaction action; the selected interaction then changes how other agents respond and therefore what evidence becomes available for subsequent inference. Established models of multi-agent decision making identify these two computations as fundamental components of interaction, motivating us to ask whether current LLM agents can reliably perform them together (Riemer et al., 2025).

We study this question in **incomplete-information, mixed-incentive coordination**. This setting is important both structurally and practically. Incomplete information makes relevant properties of other decision makers latent, creating a genuine inference problem; mixed incentives make those properties consequential for action selection, because an interaction that is beneficial under one counterpart objective may be ineffective or harmful under another. Their combination therefore preserves both sides of the interactive decision problem: the agent must infer whom it is interacting with and reason about how that inference changes the value of its own actions. These conditions arise naturally when agents act on behalf of distinct principals. Scheduling agents coordinate around privately held constraints and individual disruption costs (Zou et al., 2026); strategic negotiation agents pursue asymmetric private objectives while forming temporary alliances (O’Neill et al., 2026); and agents representing separate economic principals exhibit state-dependent strategic behavior over repeated commercial interactions (Li et al., 2026b). We therefore use incomplete-information, mixed-incentive interaction not as an attempt to reproduce every form of MAS, but as a controlled and practically meaningful realization of the interactive-decision structure above.

Having identified the capabilities of interest, the next challenge is **how to diagnose and train them**. Games are an attractive substrate because their information structure, utilities, legal actions, and outcomes can be controlled and verified. Recent work has consequently used multi-agent games and

self-play to train strategic or transferable reasoning in LLMs (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b). However, these environments are generally designed to optimize and evaluate game-level behavior rather than to provide separate supervision for interactive-state inference and the planning that follows from it. Scalable self-play often relies on compact competitive or cooperative games that do not simultaneously require incomplete information and mixed incentives, while richer social games can combine deduction, communication, deception, coordination, memory, and long-horizon execution within the same trajectory (Light et al., 2023). Both are valuable training paradigms, but neither is designed around the diagnostic granularity required here: for a particular interaction, **what should the agent infer, and what should it do given that inference?**

We construct a controlled game environment to make both quantities explicit. Starting from commitment-based multi-party negotiation (Benac et al., 2026), we introduce private player preferences and incomplete information. Players assign different values to public goals and interact through consequential offers, responses, and binding commitments. Their actions therefore serve two roles at once: they alter future strategic opportunities and provide evidence about the latent preferences that generated them. The environment is deliberately compact. Its purpose is not to maximize surface realism, but to preserve the structural ingredients of incomplete-information, mixed-incentive interaction while remaining small enough for exact filtering over latent agent states, oracle partner policies, and counterfactual evaluation of alternative interactions. This gives us a controlled realization of the classical interactive-decision problem in which both inference and planning can be inspected directly.

We first use this environment to diagnose current LLM agents. At the same decision point, we can separately test whether a model’s inference is supported by the interaction history and whether its planned interaction is appropriate under that inference. This exposes errors that final game reward alone cannot distinguish and allows us to examine how inference and planning interact. We then construct **strategic game slices** around such decision points. Each slice retains the interaction history, latent state, incentive structure, and relevant continuation of the original game, but provides oracle targets for both what can be inferred and how alternative interactions should be valued. The purpose of these slices is not to claim that shortened-game training is generally superior to full-game self-play; they provide a controlled unit for supplying detailed supervision to the specific computations revealed by our diagnosis.

Finally, we train LLM agents using this supervision and ask whether the resulting improvement extends beyond the states on which it was provided. We evaluate generalization to structurally unseen configurations of the training game and transfer to external multi-agent environments with different interaction mechanics. Our aim is therefore broader than producing a stronger negotiation agent: **we test whether targeted game supervision can improve interaction reasoning that remains useful when the surface game changes.**

Our contributions are threefold:

- **A controlled diagnosis of interactive reasoning in LLM-based MAS.** Grounded in classical interactive decision making, we study interactive-state inference and interaction planning in incomplete-information, mixed-incentive coordination, and separately measure failures in the two while retaining their sequential dependence.
- **A compact game construction for verifiable social-reasoning supervision.** We show how private preferences, consequential commitments, and information-revealing actions can instantiate a practically relevant interactive-decision problem while providing exact latent-state targets and counterfactual action values.
- **Targeted game supervision and transfer.** We construct strategic game slices around diagnosed interaction decisions, use their oracle structure to supervise inference and planning, and test whether the resulting capabilities generalize to unseen strategic configurations and external multi-agent environments.

2 RELATED WORKS

Classical multi-agent decision making couples inference about other agents with downstream action selection, from opponent models and interactive beliefs (Carmel & Markovitch, 1996; Gmytrasiewicz & Doshi, 2005; Albrecht & Ramamoorthy, 2013) to belief-space coordination, inference

from actions, and opponent-aware planning (Macindoe et al., 2012; Wu et al., 2021; Foerster et al., 2019; Huang et al., 2024). For LLM agents, explicit mental-state or behavioral representations support collaboration, adaptation, trust tracking, and planning (Li et al., 2023; Cross et al., 2025; Jha et al., 2026; Mu et al., 2026; Zhou et al., 2026; Feng et al., 2026; Goel et al., 2026; Hwang et al., 2026). However, accurate mental-state prediction need not yield useful behavior (Riemer et al., 2025; Sobotka et al., 2026).

Recent MAS benchmarks expose strategic failures under private information, mixed incentives, and imperfect communication. Commitment-based negotiation, self-negotiated contracts, and multi-party conquest model consequential agreements and coalition dynamics (Benac et al., 2026; Wyse et al., 2026; O’Neill et al., 2026); CalBench, SOTOPIA-TOM, and SNEAK make information disclosure and privacy strategically consequential (Zou et al., 2026; YS et al., 2026; Avsian & Heck, 2026). Broader evaluations cover collaboration, controlled interaction, long-horizon commerce, and communication barriers (Zhu et al., 2025; Chen et al., 2026; Li et al., 2026b; Xuan et al., 2026). Diagnostic work further finds failures in distributed information pooling, distributed computation, dynamic grounding, and belief-sensitive negotiation (Li et al., 2026a; Zhang et al., 2026b; Yao et al., 2026; Zhang et al., 2026a; Agashe et al., 2025); these complement broad failure taxonomies and accounts of long-lived peer interaction (Cemri et al., 2025; Anthropic, 2026).

Game-based training has been used to induce transferable strategic and social reasoning through self-play, multi-game reinforcement learning, and social reasoning supervision (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b; Hua et al., 2026; He et al., 2026a; Wu et al., 2026). Cross-environment cooperation and Other-Play study generalization across tasks and partners (Jha et al., 2025; Hu et al., 2020). Curriculum work selects terminal-proximal or high-learning-value subgames (West et al., 2020; Chen et al., 2024), while LLM curricula can also induce undesirable strategic abstractions (Madmoun & Lahlou, 2026). Similarity-based inference has also been proposed as a route to cooperation among foundation-model agents (Meulemans et al., 2026).

3 PRELIMINARIES

3.1 FROM INTERACTIVE UNCERTAINTY TO TWO CAPABILITIES

Coordination with private information is naturally an interactive partially observable decision problem. For agent i , an I-POMDP is

$$\mathcal{I}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle, \quad IS_i = S \times \prod_{j \neq i} M_j, \quad (1)$$

where S is the physical state, M_j models agent j , A is the joint action space, and T_i, O_i, R_i specify transition, observation, and reward (Gmytrasiewicz & Doshi, 2005). Uncertainty therefore covers not only the world but also others’ information, preferences, and policies. This view underlies opponent modelling and belief-space coordination (Carmel & Markovitch, 1996; Huang et al., 2024; Goel et al., 2026); the observation–belief–action links nevertheless remain fragile for LLM agents (Sobotka et al., 2026; Jha et al., 2026).

Let $b_t^i \in \Delta(IS_i)$ be agent i ’s belief after public history h_t . An observation updates it as $b_{t+1}^i = \tau_i(b_t^i, a_t^i, o_{t+1}^i)$, while an action is selected by belief-conditioned planning,

$$a_t^{i,*} \in \arg \max_{a \in A_i} \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} R_i(x_k, a_k) \mid b_t^i, a_t^i = a, \pi_{-i} \right]. \quad (2)$$

We call these computations *partner inference* (B), which maintains the part of b_t^i predictive of others, and *partner-conditioned planning* (P), which evaluates actions under that belief. They form the loop $B_t \rightarrow P_t \rightarrow a_t^i \rightarrow o_{t+1}^i \rightarrow B_{t+1}$: belief changes the best interaction, while the chosen interaction determines the next evidence. This is a functional decomposition, not a claim about separate neural modules.

3.2 BENAC-I: NEGOTIATION WITH PRIVATE PREFERENCES

To expose both computations while retaining exact supervision, we extend the commitment-based negotiation game BENAC (Benac et al., 2026). We call the resulting incomplete-information variant

BENAC-I. An instance is

$$\mathcal{E} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \delta, \rho, C_0, \mathcal{L}, \{\Theta_i\}_{i \in \mathcal{N}}, \mu_0, \mathbf{q} \rangle. \quad (3)$$

Here $\mathcal{N}$ is the player set, K_i counts i 's binary commitment coordinates, and each public goal $g \in \mathcal{G}$ has requirements $\Gamma_g \subseteq \{(i, k) : k < K_i\}$ and mode $\delta_g \in \{\text{binary}, \text{linear}\}$. The proposer schedule ρ and legal-action relation $\mathcal{L}$ govern play from the initially empty, irreversible commitment state C_0 . Each player privately observes a preference type $v_i \in \Theta_i \subseteq \{-1, 0, +1\}^{|\mathcal{G}|}$, drawn from common prior μ_0 ; entries mean AVOID, NEUTRAL, or WANT. Vector $\mathbf{q}$ records remaining investigation budgets.

For commitment state C , goal satisfaction and terminal utility are

$$S_g(C) = \begin{cases} \mathbf{1}[c_{ik} = 1, \forall (i, k) \in \Gamma_g], & \delta_g = \text{binary}, \\ |\Gamma_g|^{-1} \sum_{(i, k) \in \Gamma_g} c_{ik}, & \delta_g = \text{linear}, \end{cases} \quad U_i(C; v_i) = \sum_g v_{ig} S_g(C). \quad (4)$$

On a proposal turn, a player may pass, offer new commitments (optionally as two alternatives), or use INVESTIGATE(j, g) when $q_i > 0$. Acceptance irreversibly binds both sides' proposed additions; rejection binds none; choosing from a two-offer proposal binds only the selected option. Investigation instead consumes the turn and one unit of q_i , leaves C unchanged, and privately reveals v_{jg} ; the target, but not the answer, is public. Negotiation therefore supplies indirect evidence and investigation supplies costly direct evidence within the same action space.

Reference teacher and labels. The teacher is an information-limited policy profile, not an omniscient player. At an information set $I_j = (h, v_j, r_j)$ —public history, own preferences, and private investigation results—it chooses

$$\text{supp } \pi_j^*(\cdot \mid I_j) \subseteq \arg \max_{a \in \mathcal{L}(I_j)} \mathbb{E}_{v_{-j} \sim b_j(\cdot \mid I_j)} [U_j(C_T; v_j) \mid a, \pi_{-j}^*], \quad (5)$$

breaking own-utility ties by total utility to others and mixing over residual ties. We enumerate the finite game tree, initialize legal actions uniformly, and synchronously update information-set best responses until the whole contingent profile is stable. We retain it only after a unilateral-deviation check; cycles and resource-limit failures yield no label. This defines a reproducible reference, not a unique or general-sum Nash equilibrium.

The same profile supplies both labels. An observed action gives the exact finite-support update $b_{i+1}^i(v_{-i}) \propto b_i^i(v_{-i}) \pi_j^*(u_i^j \mid I_i^j, v_j)$, treating i 's actions as interventions and conditioning on any private investigation result; this is the B target. For P , exact continuation gives $Q_i^*(I_i, a)$ for every legal action and hence every optimal action. Both labels thus share one declared behavioral model and terminal objective.

4 DIAGNOSE

We test whether a model can infer a partner's preference (B), act under a correct judgment (P), and compose the two. We freeze legal three-player histories and intervene on the judgment and first action, which end-game reward alone cannot do.

4.1 CONTROLLED NATIVE ENDGAMES

We evaluate Qwen3-4B-Instruct-2507 at temperature zero on 36 positions. Each preassigned stratum-split cell contains six distinct source games, with strata *known*, *unknown-irrelevant*, and *unknown-relevant*. Ego knows all but one partner-goal entry, whose possibilities are {WANT, NEUTRAL, AVOID}. We retain the public history, commitments, schedule, and all legal actions. Information-limited reference partners compute an exact terminal best response to a declared continuation. At most two ego decisions are measured, making counterfactual enumeration exact.

Selection uses game mechanics, never model answers. In an unknown-relevant position, an action optimal for one supported type is strictly suboptimal for another, certifying $B \rightarrow P$. The same position has two legal first actions whose response branches differ in expected posterior entropy, certifying $P \rightarrow B$. Unknown-irrelevant positions control for uncertainty that does not change the optimal-action set. These are task certificates, not claims about internal neural modules.

Table 1: Diagnosis on 12 unknown-relevant endgames. Exactness requires the full supported set; regret is in terminal utility units.

Quantity	Result
$B \rightarrow P$: type changes an optimal action	12/12
$P \rightarrow B$: action changes evidence entropy	12/12
Mean action loss / entropy contrast	1.50 / 0.94 bits
Root partner judgment exactly correct	1/12 (8.3%)
Planning regret with oracle judgment	0.56
Positive regret / exact post-evidence belief	8/12 / 26.4%

For B , the model returns every type compatible with history; unresolved sets are correct. For P , we supply the exact judgment and measure $\max_{a'} Q_i^*(h, b^*, a') - Q_i^*(h, b^*, a)$. We then force lower- and higher-information actions, enumerate all responses, and test the next belief, removing the model’s first choice as a confound.

4.2 BOTH CAPABILITIES FAIL WHERE BOTH ARE REQUIRED

Table 1 pools the two preassigned splits for the 12 unknown-relevant positions. In all 12, partner information changes both the best current action and what can be learned next.

The exact support is recovered once in 12 roots; the model averages 0.54 false exclusions. Even with the oracle judgment, regret is positive in eight cases (mean 0.56). After oracle-selected interactions, the next belief is exact on only 26.4% of probability-weighted branches. The model therefore both misreads evidence and fails to exploit a correct reading.

Representative failure patterns. Three cases make this concrete. (i) In `native-30005-g0-d1`, a partner’s offer is compatible with all three types, but the model treats a commitment used by the goal as a declaration of WANT and collapses the support. (ii) In the same state, even when given the correct support, it proposes a two-option offer to P_1 worth 0, while an offer to P_0 is worth 1: a planning error under an oracle judgment. (iii) In `native-30040-g3-d2`, an informative interaction separates a WANT branch from a {NEUTRAL, AVOID} branch. The model correctly outputs WANT after acceptance, yet also outputs it after the diagnostic rejection, reversing the evidence on that branch.

Known preferences do not eliminate planning error, while a wrong judgment need not matter when the entry is decision-irrelevant. Our scoped conclusion is that both B and P fail in states that couple them, and either failure can block a higher-value interaction. This motivates separate supervision of the two targets and explicit evaluation of their composition.

5 TRAINING

6 TEST AND TRANSFER

7 DISCUSSION

8 CONCLUSION

AI USE STATEMENT

We used generative AI tools to assist with literature discovery, reference formatting, and drafting and editing portions of this manuscript. All AI-assisted text and bibliographic metadata were reviewed by the authors. The authors take responsibility for the final content of the paper, including its claims, experimental artifacts, and citations.

ETHICS STATEMENT

This work studies LLM-agent interaction in a synthetic, simulated negotiation environment and does not involve human participants or personal data. The environment includes mixed incentives and strategic information disclosure, which can surface behaviors such as manipulation or unhelpful information sharing. We use these behaviors only for controlled evaluation and training of coordination capabilities. We document the game rules, oracle policies, and limitations so that the resulting methods are not interpreted as deployment policies for real-world autonomous agents.

REPRODUCIBILITY STATEMENT

The main text and appendix specify the BENAC-P game rules, information structure, oracle belief filtering, and reference planning procedure. The appendix reports the diagnostic positions, prompts, evaluated model version, decoding configuration, controls, aggregation procedure, random seed, and provenance information used for the reported results. These details are intended to make the diagnostic suite and its measurements reproducible.

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A NATIVE DIAGNOSTIC PROTOCOL AND SUPPLEMENTARY RESULTS

Positions and oracle selection. Each frozen position contains its original game specification, independent per-player preference catalogues, ego’s preference row, a queried partner–goal pair and a replayable legal history. The ten positions have three players, two binary commitments per player, eight public goals and two scheduled proposal turns per player. At most three original proposal turns remain at the root. No preparation, assessment or automatic approval action is introduced. The active partner policy uses public history and its own preferences; its terminal best response is to the declared fixed continuation, not to an omniscient or equilibrium opponent.

For the judgment-sensitive cohort, the oracle evaluates all legal actions under each still-supported partner type. Admission requires an action optimal under one type with positive regret under another. This witnesses a consequential dependence on partner understanding, while allowing actions that are optimal under multiple types. It neither requires the model’s judgment to be wrong nor guarantees a change when that particular judgment is repaired. For the action–evidence cohort, native enumeration selects the first pair of legal actions with different expected posterior entropy and a nonterminal next ego decision on every branch. “Higher” and “lower” information refer only to this pair, not global information extrema. The utility-optimal reference action is selected independently. A positive downstream value from updating a frozen prior is not an admission condition.

A.1 WORKED MENU EXAMPLE

In `native-30040-g6-d2`, let x_i denote the already-bound $P_i.A_0$ and y_i denote $P_i.A_1$. Initially all x_i are true and all y_i are false. Table 3 gives the complete game; only v , P_2 ’s preference for G_6 , varies across the three compatible types. The public prefix is P_0 pass; a P_1 menu choosing between binding $(P_1.A_0, P_2.A_0)$ and $(P_1.A_1, P_2.A_0)$, with P_2 choosing the former; P_2 offering to add $P_0.A_0$, accepted by P_0 ; and P_2 offering to add $P_1.A_1$, rejected by ego. The resulting state is the example’s root.

Table 2: Frozen native positions. D and C retain the source-seed discovery and confirmation labels used during development; C is not an untouched test set. Each direction has three games in each partition.

Position identifier	Partition	$B \rightarrow P$	$P \rightarrow B$
native-30003-g6-d2	C	yes	
native-30005-g0-d1	C	yes	
native-30009-g2-d2	C	yes	
native-30013-g6-d3	C		yes
native-30016-g5-d1	D	yes	yes
native-30018-g5-d2	D	yes	yes
native-30019-g2-d1	C		yes
native-30020-g5-d1	D	yes	
native-30041-g6-d1	C		yes
native-30040-g6-d2	D		yes

Table 3: Complete goal and preference specification of the worked example. Entries +1, 0 and −1 mean WANT, NEUTRAL and AVOID. Here $v \in \{+1, 0, -1\}$ is hidden from ego.

Goal	Required commitments	P_0	P_1 (ego)	P_2
G_0	$x_0 \wedge y_1$	0	+1	0
G_1	$x_0 \wedge y_1 \wedge x_2$	−1	+1	+1
G_2	$y_0 \wedge x_1 \wedge y_2$	+1	−1	−1
G_3	$y_0 \wedge y_1 \wedge y_2$	0	+1	−1
G_4	$x_0 \wedge x_1$	+1	+1	0
G_5	$x_1 \wedge y_2$	0	−1	+1
G_6	$y_0 \wedge y_2$	+1	+1	v
G_7	$y_1 \wedge x_2$	−1	−1	+1

The menu offers O_1 , adding only y_2 , and O_2 , adding y_1 and y_2 . Exact terminal values for P_2 , evaluated under its declared fixed reference continuation, are

v	REJECT	CHOOSE_1	CHOOSE_2	oracle response
+1	1	1	2	CHOOSE_2
0	0	1	1	CHOOSE_2
−1	0	1	0	CHOOSE_1

For the neutral type, the immediate-reference tie-break selects O_2 : it immediately satisfies P_2 ’s wanted G_1, G_5, G_7 , whereas O_1 satisfies only G_5 . Thus CHOOSE_1 identifies AVOID, and CHOOSE_2 leaves WANT/NEUTRAL. Offering O_1 alone elicits acceptance from all three types. The comparison illustrates how the menu changes the evidence channel without changing the legal game protocol.

After either menu choice, the actual active P_0 offers to add y_0 while asking ego for no additional commitment. After O_1 , ego’s terminal values for reject and accept are both 0. After O_2 , they are 1 and 2 respectively. The difference follows from the bound commitments: adding y_0 after O_1 activates an avoided G_2 and a wanted G_6 , which cancel; after O_2 it also activates the wanted G_3 . The observed model judgment after the O_1 menu branch is {WANT, NEUTRAL}, excluding the only supported type. This example demonstrates the action–evidence contrast and an inference error; its between-action utility difference also reflects physical commitments. It is one of the six action–evidence cases, not an additional replication.

A.2 MEASUREMENT AND EXECUTION DETAILS

Queries and interventions. Let $B^*(s, H)$ denote the evidence-supported preference set and $\hat{B}(s, H)$ the model’s reported set. Both paired planning queries receive the same state s , complete public history H , initial possibilities and legal actions. Only the explicit judgment differs. The belief-generation scratchpad is not passed to either fresh planning context. For a supplied judgment B , write the model action as $\hat{a}(B)$ and let $Q^*(a)$ be its exact expected terminal value under the evidence-supported reference judgment. We report

$$\text{Reg}(a) = \max_{a'} Q^*(a') - Q^*(a), \quad \Delta_B = \text{Reg}(\hat{a}(\hat{B})) - \text{Reg}(\hat{a}(B^*)). \quad (6)$$

Positive Δ_B means beneficial repair; either sign can accompany an action change. Exact-set B accuracy requires every supported preference and no unsupported preference. Reference beliefs are uniform over compatible types; all probabilities and action values are computed in the backend.

For forced action a , let E_a be the subsequent sequence of observed partner actions before the next ego decision, and let Z be the uncertain partner type. The channel statistic is $\mathbb{E}[H(Z | H, a, E_a)]$ in bits, with every branch weighted by its compatible-type mass. The information gap subtracts this statistic for the higher-information arm from that for the lower-information arm. Model exactness is scored against the appropriate support on each branch, then averaged with the same weights. Terminal branches in other action arms require only a judgment; no fictitious continuation action is added. After the second measured ego action, the reference planner controls the remaining continuation.

Execution and semantic controls. We use Qwen3-4B-Instruct-2507 with temperature zero, served through vLLM 0.28 and the Hermes tool-call parser. The prompt requests brief reasoning with soft targets of 120 words for judgments and 240 for actions. Each response exposes one submission tool with automatic tool choice and a 1,024-token completion cap. Only truncation permits one additional 128-token submission request, without corrective feedback. All 127 formal tasks were valid on their first request; no finalization was needed. All contained assistant text before the tool call, averaging 131.4 whitespace-delimited words and 230.8 completion tokens per task (29,308 completion tokens in total). These traces are observable explanations, not a validated measure of the model’s internal computation.

Four separate empty-history controls supply singleton WANT, NEUTRAL, AVOID, or the unresolved three-label set. All four returned valid submissions, but the WANT control incorrectly returned all three labels. Its explanation incorrectly described the supplied initial set as containing all three values. The run initially stopped under the semantic gate and then continued under a protocol-validity gate, reusing all four cached responses, including this error. Neither prompts nor inference budgets were changed, and incorrect semantic answers were not retried. The controls consumed 491 completion tokens in four requests, separately from formal-task counts. This failure limits attribution of B errors: some can reflect reading or prior-grounding failures rather than strategic inference alone.

Aggregation and partition results. Panel A of Table 1 counts the six certified positions in each cohort. These counts summarize admission criteria, not six successful model interventions. The judgment-sensitive certificate uses two compatible type-conditioned singleton judgments at the same public state and history; it does not claim that the actual history identifies either singleton. The model’s oracle-judgment input instead contains the full evidence-supported set. Panel B reports model errors under the controlled inputs, separately from these task-level certificates.

The main table combines the two development partitions *within* each six-game cohort. It never pools the overlapping cohorts as twelve independent games. Branches are weighted within games before computing game means. The 95% intervals use 2,000 percentile-bootstrap resamples of source games with random seed zero. They describe variation in this selected small set, not failure prevalence across random games or an untouched confirmatory test. The 0.918-bit channel gap is identical in all six selected pairs; its absence of between-game variation is a property of these examples, not a precision claim about other environments.

Table 4: Results by retained development partition, three games per cell. Regret and judgment repair use the cohort roots. Forced-arm B accuracy is branch-weighted. Judgment replacement changes one action in C and two in D.

Measure	D	C
<i>Judgment-sensitive cohort</i>		
B exact	0.0%	33.3%
P regret with reference judgment	0.389	0.500
Judgment repair: reduction in regret	−0.222	0.000
<i>Action-evidence cohort</i>		
Root P regret with reference judgment	0.667	1.333
B exact, lower-information action	0.0%	33.3%
B exact, higher-information action	11.1%	33.3%

Task consequences and secondary decompositions. The following joint-response measurements supplement the oracle certificates and interface-specific error measurements; their sign is not an admission criterion or a test of whether the task dependency exists. Across the six judgment-sensitive cases, mean model-planner regret reduction is -0.111 $[-0.333, 0.000]$. Across the six action-evidence cases, the paired difference in B exactness is 5.6 $[-38.9, 44.4]$ percentage points. Neither quantity establishes a positive repair benefit.

In the judgment-sensitive cohort, replacing the explicit judgment changes `native-30005-g0-d1` from a menu to an offer and `native-30016-g5-d1` from a menu to pass, with no change in expected utility. At `native-30018-g5-d2`, it changes pass to an offer and increases regret from zero to $2/3$. These are observed effects of the supplied judgment on the same planner, not evidence that erroneous judgments are generally preferable. The other three root actions are unchanged.

For the forced action pairs, the higher-minus-lower reference utility contrast is 0.444 $[-0.222, 1.111]$; with the model choosing the second action, it is -0.222 $[-1.333, 0.833]$. These quantities include physical commitment changes and do not identify information-mediated utility effects. The auxiliary chooser/updater decomposition has ten complete tables, but replacing the updater changes reference-continuation utility by zero in both the utility-optimal and model-selected first-action arms in every game. We do not use these zeros as evidence that inference is irrelevant, or claim a positive updater-repair benefit. One root reference-planner score under a wrong model judgment is undefined because the assumed type contradicts the public history; that secondary score is marked unavailable, not filled with zero. Direct model action regret and the paired model-planner intervention remain defined.

Relation to the preceding native diagnosis. An earlier native run covered 36 positions from 26 source games, with known, unknown/relevant and unknown/irrelevant conditions. Its six designated confirmation unknown/relevant primary cases had B exact agreement 16.7% $[0.0, 50.0]\%$ and reference-judgment P regret 0.667 $[0.250, 1.083]$. Those primary cases mixed root and post-interaction decisions. That run produced 301 valid submissions out of 302 tasks and passed all four semantic controls. We retain it as preceding B/P evidence, without pooling it with this supplement: nine positions are reused, and the additional position is another query/history from an already represented source game. The earlier restricted three-stage protocol, which changed action structure and coupled partner types, is not the source of any results reported here.